

对Rinaldi等人(2016)研究结果的 计算可复现性检验

援手有序：视运动惯例对数字与非数字序列空间表征的组织作用

汇报人：凌幸

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Rinaldi, L., Di Luca, S., Henik, A., & Girelli, L. (2016). A helping hand putting in order: Visuomotor routines organize numerical and non-numerical sequences in space. *Cognition*, 152, 40–52. <https://doi.org/10.1016/j.cognition.2016.03.003>

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01 研究背景与争议

✿ 经典现象：SNARC效应 (Spatial Numerical Association of Response Codes)

- 西方被试：左手对小数反应更快，右手对大数反应更快 (Dehaene et al., 1993)。
- 心理数字线 (MNL)：数字在心理空间内从左到右排列。

📖 (一) 文化观：阅读方向 (Reading Direction, RD)

- 读写时的扫描方向会塑造内部表征的空间偏好。
- 希伯来语人群（从右到左书写），SNARC效应出现反转或显著弱化。

👉 (二) 具身观：手指计数 (Finger Counting, FC)

- 儿童期用手指计数的具身经验是空间映射的根源。
- 原始的认知表征，优先级可能高于后天习得的阅读习惯。

01 研究问题

核心问题：手指计数习惯（FC）与阅读方向（RD）作为两种基础的空间经验，是如何作用于数字与非数字序列的空间表征？



实验1：数字表征溯源

验证数字的空间映射是遵循心理数字线（MNL），还是手指计数习惯（FC）。



实验2：跨语言稳定性检验

探究 FC 映射是否独立于阅读方向，且具有跨语言稳定性？



实验3：非数字序列的拓展

探索 FC 是否同样影响有顺序的非数字序列（星期）？

01 研究方法



被试：双向阅读者

- 以色列双语右利手学生
- ✓ 母语希伯来语（右→左）
- ✓ 第二语言英语（左→右）



统计方法：SPSS

- RM-ANOVA
- One-way ANOVA
- Scheffé 事后比较



自变量

- **Direction:** TL、LT
TL: 拇指→小指（匹配FC方向）
LT: 小指→拇指（不匹配FC方向）
- **Posture:** Prone (朝下)、Supine (朝上)
- **Language:** English、Hebrew



因变量

- 反应时
 - ✓ 正确试次
 - ✓ 数据校正
- 错误率：arcsin转换

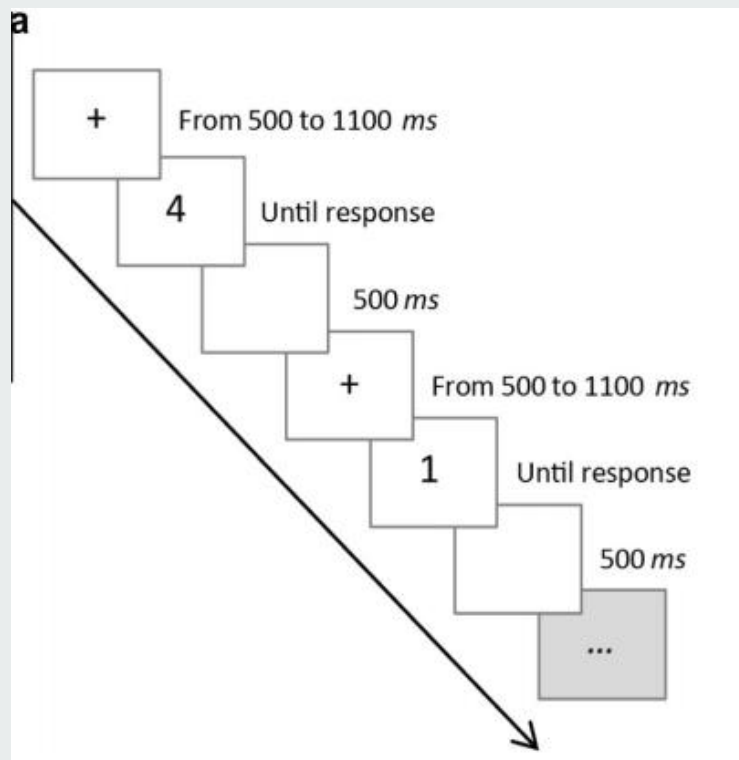
01 研究方法

实验	刺激材料	被试(有效样本)	核心自变量
实验 1	阿拉伯数字 1-5	20 (18)	2 × 2 被试内 Direction × Posture
实验 2	英文、希伯来文数字词	24 (21)	2 × 2 × 2 被试内 Direction × Posture × Language
实验 3	英文、希伯来文星期词 (以色列工作周)	22 (20)	2 × 2 × 2 被试内 Direction × Posture × Language

01 实验范式：手指映射相容性任务 + 基线任务

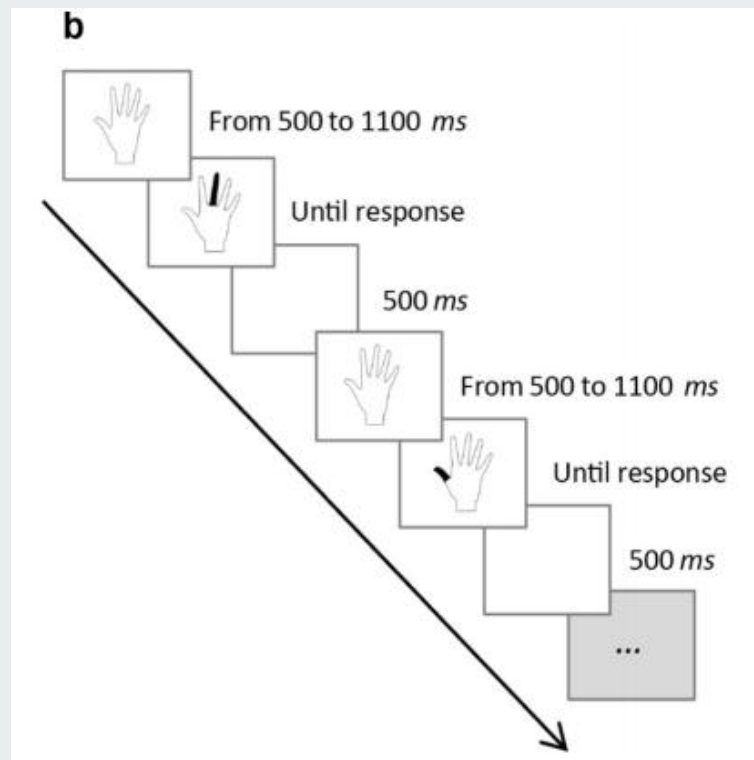
01 手指映射相容性任务

屏幕呈现数字或星期词序列刺激，被试按既定规则将刺激映射至右手五指按键，核心指标为经过基线校正后的反应时（RT）。



02 基线任务

单独测试右手五指在掌心向上/向下时的基础反应速度，建立基准参照，用于后续校正RT数据，消除手指灵活性与姿势的先天速度差异。



01 实验结果

实验	主要结果	核心结论
实验1 基础数字映射验证	• Direction主效应显著, $F(1,17) = 32.65, p < .001$ • Posture主效应和两因素交互不显著	• TL方向的反应显著快于LT方向 • 数字的空间映射遵循 FC 的方向规律
实验2 映射的跨语言稳定性	• Direction主效应与Language主效应均显著 (p均< .001) • Direction与Posture存在显著交互 ($p<.01$)	• FC 优势跨语言稳定存在, 且母语优势 • FC 优势效应受身体姿势的调节
实验3 非数字序列映射	• 三因素交互显著, $F(1,19) = 16.69, p < .002$ • FC 与 RD 同时相容时, 被试反应最快	• FC 与 RD 的相容性共同影响反应时 • 体现了双重编码的促进作用

✦ 结论：具身认知——数字的空间表征主要依赖手指计数；非数字序列同时依赖手指计数+阅读方向。

02 复现方法概述

数据来源

OSF (<https://osf.io/8btme/>)

样本量

与原文献一致
(N=18/21/20)

复现工具

- R 4.6.0 (原文使用SPSS)
- 包: tidyverse, afex, emmeans, lme4, ggplot2

复现步骤	目的
① 原始数据预处理	筛选正确试次; 剔除 $M \pm 2SD$ 极端值
② 计算校正RT	手指校正(finger Δ) + 姿势校正(posture Δ)
③ RM-ANOVA (原方法)	检验原文献结果是否可复现
④ LMM (创新方法)	验证结果对不同分析方法的稳健性

02 原始数据预处理

方案	分组方式	剔除时机	MAE (ms)	RMSE (ms)	最大绝对差 (ms)
1	group(Subject)	correct→±2SD	66.06	94.60	336.93
2	group(Subject)	±2SD→correct	66.00	94.26	336.93
3	group(Subj,Post,Finger)	correct→±2SD	80.53	125.51	914.76
4	group(Subj,Post,Finger)	±2SD→correct	79.60	124.35	914.76
5	group(overall)	correct→±2SD	71.06	102.06	408.04
6	group(overall)	±2SD→correct	70.32	100.78	408.04

MAE(Mean Absolute Error): 各方案校正RT与作者校正RT之间的平均绝对差异

RMSE(Root Mean Square Error): 平方差异平均后开方

02 数据预处理

● 工具函数:

- 剔除极端值
- 计算finger Δ
- 计算posture Δ
- 校正RT

```
# 按分组剔除  $M \pm 2SD$  之外的试次
remove_outliers <- function(data, ..., rt_col = "RT") {
  grps <- rlang::quos(...)
  data %>%
    group_by(!!!grps) %>%
    mutate(.lo = mean(.data[[rt_col]], na.rm = TRUE) - 2 * sd(.data[[rt_col]], na.rm = TRUE),
           .hi = mean(.data[[rt_col]], na.rm = TRUE) + 2 * sd(.data[[rt_col]], na.rm = TRUE)) %>%
    filter(.data[[rt_col]] >= .lo & .data[[rt_col]] <= .hi) %>%
    ungroup() %>%
    select(-starts_with("."))
}

# 手指映射表
finger_map <- c("b" = "Thumb", "h" = "Index", "j" = "Middle",
               "k" = "Ring", "l" = "Little")

# 计算 finger_D: 各手指RT - 该姿势下最快手指RT
compute_finger_D <- function(b1) {
  means <- b1 %>%
    group_by(Subject, Posture, Finger) %>%
    summarise(m = mean(RT), .groups = "drop")
  fastest <- means %>%
    group_by(Subject, Posture) %>%
    slice_min(m, n = 1, with_ties = FALSE) %>%
    transmute(Subject, Posture, fast = m)
  means %>%
    left_join(fastest, by = c("Subject", "Posture")) %>%
    mutate(finger_D = m - fast) %>%
    select(Subject, Posture, Finger, finger_D)
}

# 计算 posture_D: Supine_mean - Prone_mean
compute_posture_D <- function(b1) {
  b1 %>%
    group_by(Subject, Posture, Finger) %>%
    summarise(pm = mean(RT), .groups = "drop") %>%
    pivot_wider(names_from = Posture, values_from = pm) %>%
    mutate(posture_D = Supine - Prone) %>%
    select(Subject, Finger, posture_D)
}

# 应用校正: finger_D + posture_D(仅Supine)
correct_fm <- function(fm_data, finger_D, posture_D, mapping_posture_vec) {
  fm_data %>%
    mutate(Finger = recode(CorrectResponse, !!!finger_map),
           Posture = recode(Mapping, !!!mapping_posture_vec)) %>%
    left_join(finger_D, by = c("Subject", "Posture", "Finger")) %>%
    left_join(posture_D, by = c("Subject", "Finger")) %>%
    mutate(corrected_RT = RT - finger_D,
           corrected_RT = if_else(Posture == "Supine",
                                  corrected_RT - posture_D, corrected_RT))
}
```

02 错误率

- 描述统计：完全复现
- RM-ANOVA：部分复现

ErrorRate	实验1		实验2		实验3	
	原文献	复现结果	原文献	复现结果	原文献	复现结果
A	3.2%	3.2%	3.1%	3.1%	3.4%	3.4%
B	3.3%	3.3%	3.9%	3.9%	2.9%	2.9%
C	2.8%	2.8%	4.1%	4.1%	4.2%	4.2%
D	3.1%	3.1%	2.5%	2.5%	4.4%	4.4%
E			5.2%	5.2%	3.0%	3.0%
F			4.5%	4.5%	2.9%	2.9%
G			2.8%	2.8%	3.5%	3.5%
H			4%	4.0%	3.3%	3.3%

```
cat("\n--- 错误率 (各Mapping) ---\n")
e1 <- fm1_raw %>%
  group_by(Mapping) %>%
  summarise(ErrorRate = sprintf("%.1f%%", 100 * mean(Accuracy == 0)), .groups = "drop")
print(e1)

cat("\n--- 错误率 2x2 RM-ANOVA (arcsin转换) ---\n")
e1_anova <- fm1_raw %>%
  group_by(Subject, Mapping) %>%
  summarise(p = mean(Accuracy == 0), .groups = "drop") %>%
  mutate(Subject = factor(Subject),
         Direction = factor(if_else(Mapping %in% c("A","C"), "TL", "LT"), c("TL","LT")),
         Posture = factor(if_else(Mapping %in% c("A","B"), "Prone", "Supine"), c("Prone","Supine")),
         asin_err = asin(sqrt(p)))
e1_fit <- aov_ez("Subject", "asin_err",
               e1_anova %>% group_by(Subject, Direction, Posture) %>%
               summarise(asin_err = mean(asin_err), .groups = "drop"),
               within = c("Direction", "Posture"),
               anova_table = list(es = "pes"))
print(e1_fit)
```

		<i>F</i>	<i>df</i>	<i>p</i>	η^2p	原文献报告
实验1	Direction	1.18	1, 17	.292	.065	no significant
	Posture	0.04	1, 17	.835	.003	no significant
	Direction×Posture	0.89	1, 17	.358	.050	no significant
实验2	Direction	0.83	1, 20	.373	.040	no significant
	Posture	9.15**	1, 20	.007	.314	no significant
	Language	2.96	1, 20	.101	.129	no significant
	Direction×Posture	0.02	1, 20	.877	.001	no significant
	Direction×Language	2.04	1, 20	.168	.093	no significant
	Posture×Language	6.69*	1, 20	.018	.251	no significant
	Direction×Posture×Language	5.41*	1, 20	.031	.213	no significant
实验3	Direction	1.42	1, 19	.248	.069	no significant
	Posture	1.17	1, 19	.292	.058	$F(1,19)=6.75$, $p<.05$, $n2p=.26$
	Language	1.38	1, 19	.254	.068	$F(1,19)=7.45$, $p<.05$, $n2p=.28$
	Direction×Posture	0.08	1, 19	.784	.004	no significant
	Direction×Language	0.05	1, 19	.827	.003	no significant
	Posture×Language	0.49	1, 19	.494	.025	no significant
	Direction×Posture×Language	0.02	1, 19	.903	<.001	no significant

02 实验1对比——描述统计

- 均值层面：偏差普遍较小，控制在6%以内。
- 离散程度：偏差较大，样本数据的波动特征与原文存在较大差异。

基线任务描述性统计									
		原文献结果		复现结果		M_δ	M_评级	SD_δ	SD_评级
		M	SD	M	SD				
Posture	Prone	414.3	51	412.07	52.09	0.54%	次要偏差	2.09%	次要偏差
	Supine	443.4	51.7	438.54	55.64	1.11%	次要偏差	7.08%	次要偏差
Finger	Thumb	430.67	47.3	407.11	40.14	5.79%	次要偏差	17.84%	主要偏差
	Index	414.3	51.8	398.84	44	3.88%	次要偏差	17.73%	主要偏差
	Middle	426.6	55.6	436.16	53.8	2.19%	次要偏差	3.35%	次要偏差
	Ring	448	57	468.06	58.53	4.29%	次要偏差	2.61%	次要偏差
	Little	424.5	50.9	416.34	51.57	1.96%	次要偏差	1.30%	次要偏差

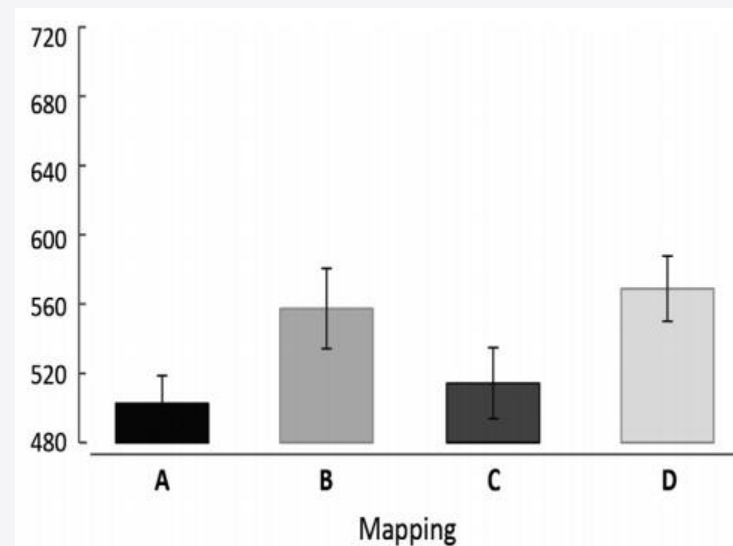
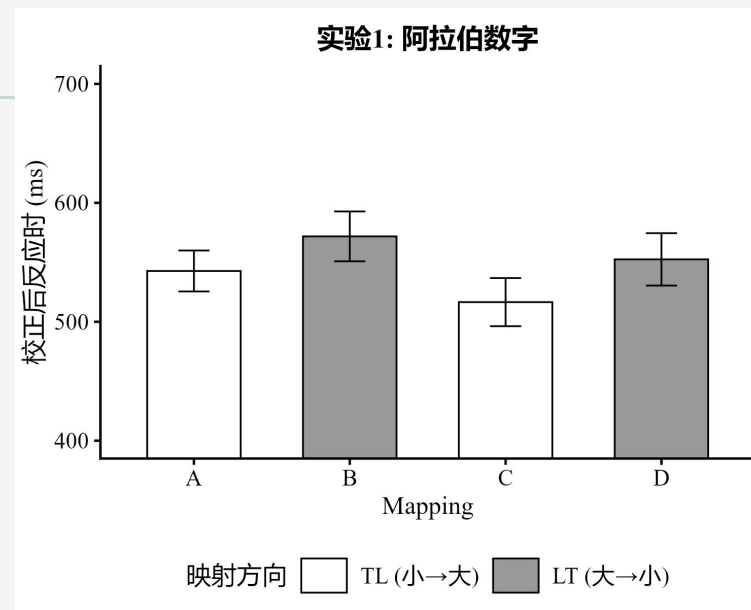
02 实验1对比——描述统计

手指映射任务描述性统计									
		原研究结果		复现结果		M_δ	M_评级	SD_δ	SD_评级
		M	SD	M	SD				
Direction	TL	508.5	125	529.61	109.67	3.99%	次要偏差	13.98%	主要偏差
	LT	563.2	133.7	562.13	126.01	0.19%	次要偏差	6.10%	次要偏差
Posture	Prone			557.24	111.78				
	Supine			534.49	125.24				
Mapping	A			542.70	96.26				
	B			571.79	124.23				
	C			516.51	120.74				
	D			552.47	127.73				
Digit	1	440.34	81.19	524.73	100.57	16.08%	主要偏差	19.27%	主要偏差
	2	548.16	93.36	538.15	110.23	1.86%	次要偏差	15.30%	主要偏差
	3	576.85	79.95	543.96	115.13	6.05%	次要偏差	30.56%	主要偏差
	4	584.45	104.20	576.33	144.05	1.41%	次要偏差	27.66%	主要偏差
	5	529.27	77.73	546.18	118.18	3.10%	次要偏差	34.23%	主要偏差

02 实验1对比——描述统计

```
cat("\n--- Exp1: 主效应柱状图 ---\n")
exp1_plot_dat <- exp1 %>%
  group_by(Direction, Posture, Subject, Mapping) %>%
  summarise(RT = mean(RT), .groups = "drop") %>%
  group_by(Direction, Posture, Mapping) %>%
  summarise(M = mean(RT), SE = sd(RT) / sqrt(n()), .groups = "drop") %>%
  mutate(Mapping = factor(Mapping),
         Direction = factor(Direction, c("TL", "LT")),
         Posture = factor(Posture, c("Prone", "Supine")))

p1_main <- ggplot(exp1_plot_dat,
                  aes(x = Mapping,
                      y = M, fill = Direction)) +
  geom_col(width = 0.6, color = "black", linewidth = 0.4) +
  geom_errorbar(aes(ymin = M - SE, ymax = M + SE), width = 0.2, linewidth = 0.4) +
  scale_fill_manual(values = c("TL" = "white", "LT" = "grey60"),
                    labels = c("TL (小→大)", "LT (大→小)")) +
  labs(x = "Mapping", y = "校正后反应时 (ms)",
       title = "实验1: 阿拉伯数字",
       fill = "映射方向") +
  theme_apo() +
  coord_cartesian(ylim = c(400, 700))
```

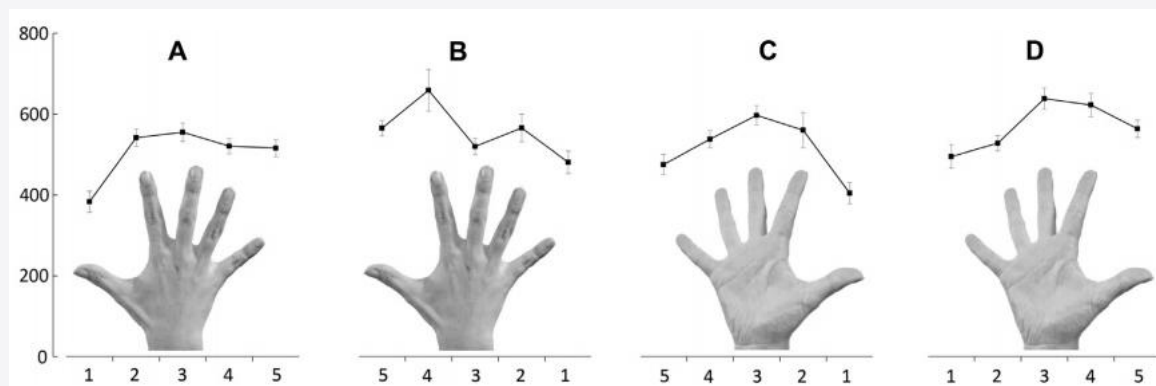
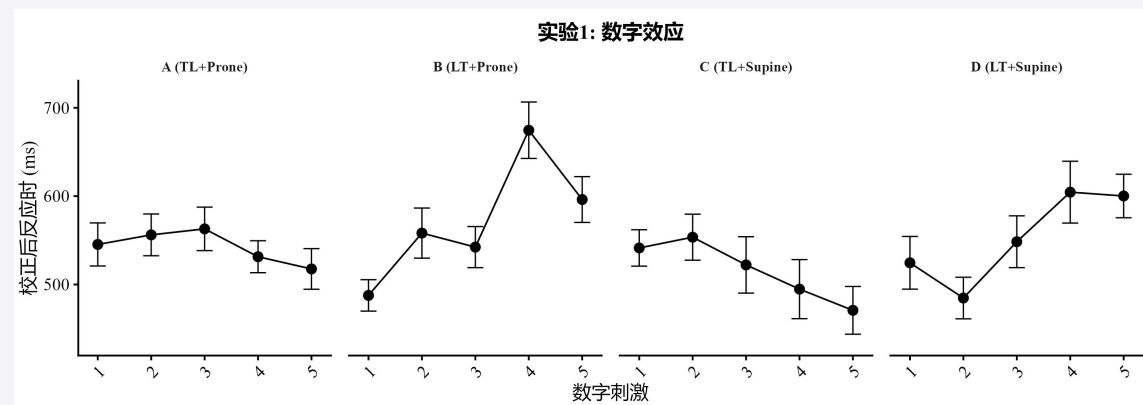


02 实验1对比——描述统计

```
cat("\n--- Exp1: 刺激效应折线图 (Mapping 分面板) ---\n")
exp1_stim_plot <- exp1 %>%
  group_by(Stimulus, Mapping, Subject) %>%
  summarise(RT = mean(RT), .groups = "drop") %>%
  group_by(Stimulus, Mapping) %>%
  summarise(M = mean(RT), SE = sd(RT) / sqrt(n()), .groups = "drop") %>%
  mutate(Panel = recode(Mapping,
    "A" = "A (TL+Prone)", "B" = "B (LT+Prone)",
    "C" = "C (TL+Supine)", "D" = "D (LT+Supine)",
    Panel = factor(Panel, c("A (TL+Prone)", "B (LT+Prone)",
      "C (TL+Supine)", "D (LT+Supine)"))))

ylim1 <- range(exp1_stim_plot$M - exp1_stim_plot$SE,
  exp1_stim_plot$M + exp1_stim_plot$SE, na.rm = TRUE) + c(-10, 10)

p1_stim <- ggplot(exp1_stim_plot,
  aes(x = Stimulus, y = M, group = 1)) +
  geom_line(linewidth = 0.5, color = "black") +
  geom_point(size = 2.5, color = "black") +
  geom_errorbar(aes(ymin = M - SE, ymax = M + SE),
    width = 0.3, linewidth = 0.4, color = "black") +
  scale_x_continuous(breaks = 1:5, labels = as.character(1:5)) +
  facet_wrap(~ Panel, ncol = 4) +
  coord_cartesian(ylim = ylim1) +
  labs(x = "数字刺激", y = "校正后反应时 (ms)",
    title = "实验1: 数字效应") +
  theme_apache() +
  theme(strip.text = element_text(size = 9),
    axis.text.x = element_text(angle = 45, hjust = 1))
```



02 实验1对比——推断统计

```
cat("--- 基线 2x5 RM-ANOVA (Posture x Finger) ---\n")
b11_agg <- b11 %>%
  group_by(Subject, Posture, Finger) %>%
  summarise(RT = mean(RT), .groups = "drop") %>%
  mutate(Subject = factor(Subject),
         Posture = factor(Posture, c("Prone", "Supine")),
         Finger = factor(Finger, c("Thumb", "Index", "Middle", "Ring", "Little")))
b11_fit <- aov_ez("Subject", "RT", b11_agg,
                 within = c("Posture", "Finger"),
                 anova_table = list(es = "pes"))
print(b11_fit)
```

基线任务推断性统计结果的比较（2×5 RM-ANOVA : Posture×Finger）					
		<i>F</i>	df	<i>p</i>	η² _p
Posture	原文献	15.48	1, 17	< .002	.49
	复现结果	14.25	1, 17	.002	.456
	评级			推论一致	
Finger	原文献	14.17	4, 68	< .001	.47
	复现结果	20.93	2.60, 44.24	< .001	.552
	评级			推论一致	
Posture×Finger	原文献			no significant	
	复现结果	3.09	3.12, 53.09	.033	.154
	评级			推论不一致	

02 实验1对比——推断统计

```
cat("\n--- 2x2 RM-ANOVA: Direction x Posture ---\n")
a1 <- aov_ez("Subject", "RT",
             exp1 %>% group_by(Subject, Direction, Posture) %>%
             summarise(RT = mean(RT), .groups = "drop"),
             within = c("Direction", "Posture"),
             anova_table = list(es = "pes"))
print(a1)
```

- 结论一致： Direction 主效应复现成功，与原文一致，即TL方向的反应速度显著更快。
- 复现差异： 效应量减小； 原文不显著的 Posture 主效应在复现中显著。

手指映射任务推断性统计结果的比较（2×2 RM-ANOVA: Direction×Posture）					
		<i>F</i>	df	<i>p</i>	η² _p
Direction	原文献	32.65	1,17	< .001	.66
	复现结果	12.25	1,17	.003	.419
	评级			推论一致	
Posture	原文献			no significant	
	复现结果	5.91	(1,17)	.026	.258
	评级			推论不一致	
Direction×Posture	原文献			no significant	
	复现结果	0.21	1,17	.651	.012
	评级			推论一致	

02 实验1对比——推断统计

```
cat("\n--- Digit One-way RM-ANOVA (1-5) ---\n")
ald <- aov_ez("Subject", "RT",
              exp1 %>% group_by(Subject, Stimulus) %>%
                summarise(RT = mean(RT), .groups = "drop") %>%
                mutate(Stimulus = factor(Stimulus)),
              within = "Stimulus",
              anova_table = list(es = "pes"))
print(ald)

cat("\n--- Digit Scheffe Post-hoc ---\n")
print(pairs(emmeans(ald, ~ Stimulus), adjust = "scheffe"))
```

One-way-ANOVA: Stimulus				
	<i>F</i>	df	<i>p</i>	η^2p
原文献	26.23	4, 68	< .001	0.61
复现结果	3.53	2.96, 50.25	.022	.172
评级	推论一致			

contrast	estimate	SE	df	<i>t</i>	<i>p</i>	原文献 <i>p</i> 值	结论
1-2	-13.42	12.50	17	-1.071	0.882	< .001	不一致
1-3	-19.24	15.40	17	-1.253	0.811	< .001	不一致
1-4	-51.60	11.80	17	-4.368	0.009	< .001	一致
1-5	-21.45	12.10	17	-1.780	0.546	< .01	不一致
2-3	-5.81	15.10	17	-0.385	0.997		一致
2-4	-38.18	9.25	17	-4.129	0.014	< .01	一致
2-5	-8.03	13.50	17	-0.597	0.985		一致
3-4	-32.37	15.60	17	-2.075	0.399		一致
3-5	-2.22	19.30	17	-0.115	1.0000		一致
4-5	30.15	15.80	17	1.903	0.483	< .05	不一致

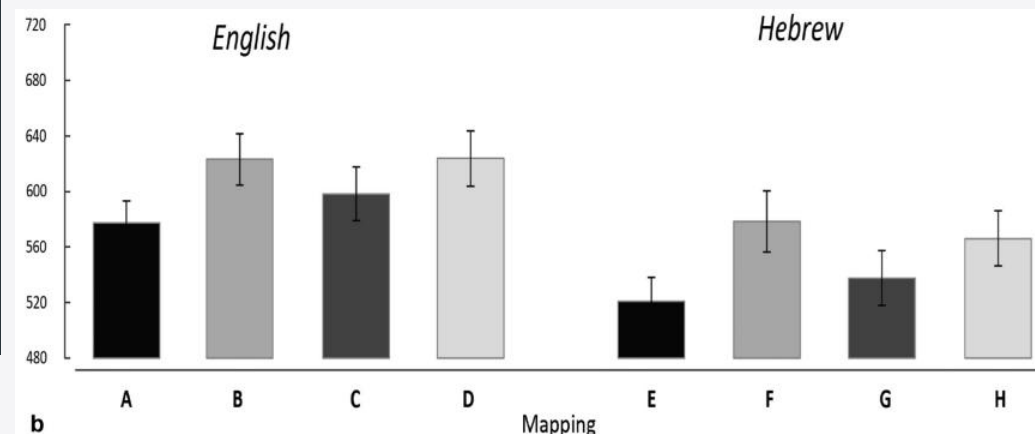
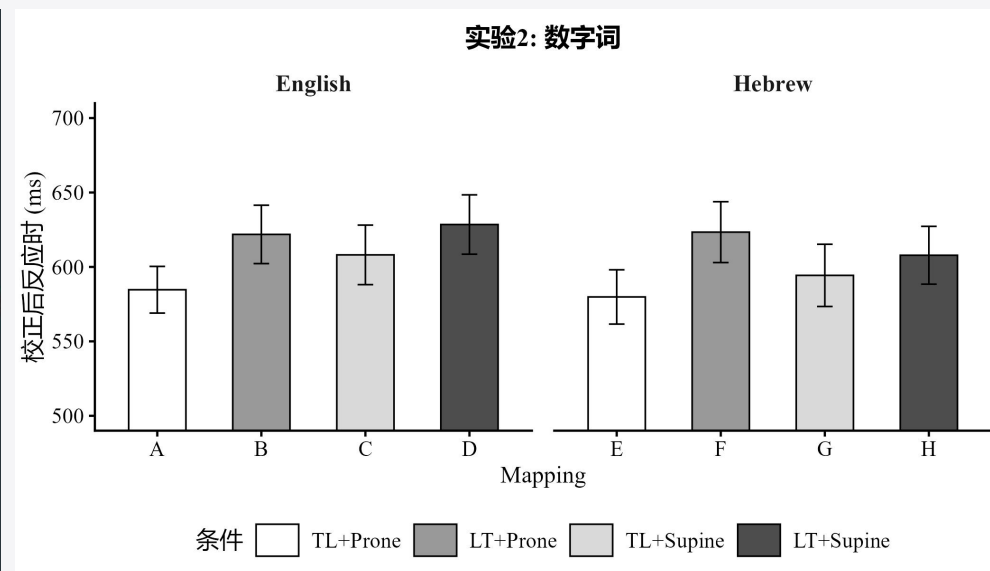
02 实验2对比——描述统计

		原文献结果		复现结果		M_δ	M_评级	SD_δ	SD_评级
		M	SD	M	SD				
Direction	TL	558.6	82.4	591.74	114.84	5.60%	次要偏差	28.25%	主要偏差
	LT	597.9	91.9	620.39	114.26	3.63%	次要偏差	19.57%	主要偏差
Language	English	605.7	84.8	610.78	115.22	0.83%	次要偏差	26.40%	主要偏差
	Hebrew	550.8	92.2	601.35	115.48	8.41%	次要偏差	20.16%	主要偏差
Mapping	A			584.68	98.10				
	B			621.85	105.51				
	C			608.10	127.65				
	D			628.49	123.67				
	E			579.84	102.78				
	F			623.38	110.42				
	G			594.33	127.22				
	H			607.84	117.22				
Number words - English	One	535.34	64.59	564.21	78.70	5.12%	次要偏差	17.93%	主要偏差
	Two	610.19	66.52	601.73	75.25	1.41%	次要偏差	11.60%	主要偏差
	Three	654.11	96.85	636.45	102.08	2.77%	次要偏差	5.12%	次要偏差
	Four	635.46	82.28	629.90	88.40	0.88%	次要偏差	6.92%	次要偏差
	Five	593.2	76.22	621.61	90.35	4.57%	次要偏差	15.64%	主要偏差
Number words - Hebrew	One	503.49	66.95	567.57	72.75	11.29%	主要偏差	7.97%	次要偏差
	Two	585.59	91.96	609.83	77.74	3.97%	次要偏差	18.29%	主要偏差
	Three	584.62	104.29	606.80	111.39	3.66%	次要偏差	6.37%	次要偏差
	Four	574.67	86.68	626.75	97.11	8.31%	次要偏差	10.74%	主要偏差
	Five	505.66	64.4	595.78	77.18	15.13%	主要偏差	16.56%	主要偏差

02 实验2对比——描述统计

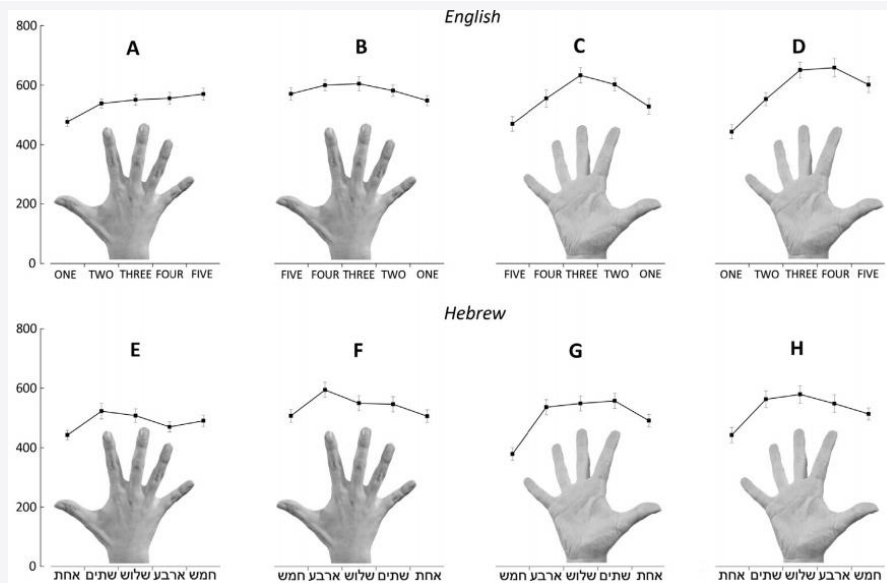
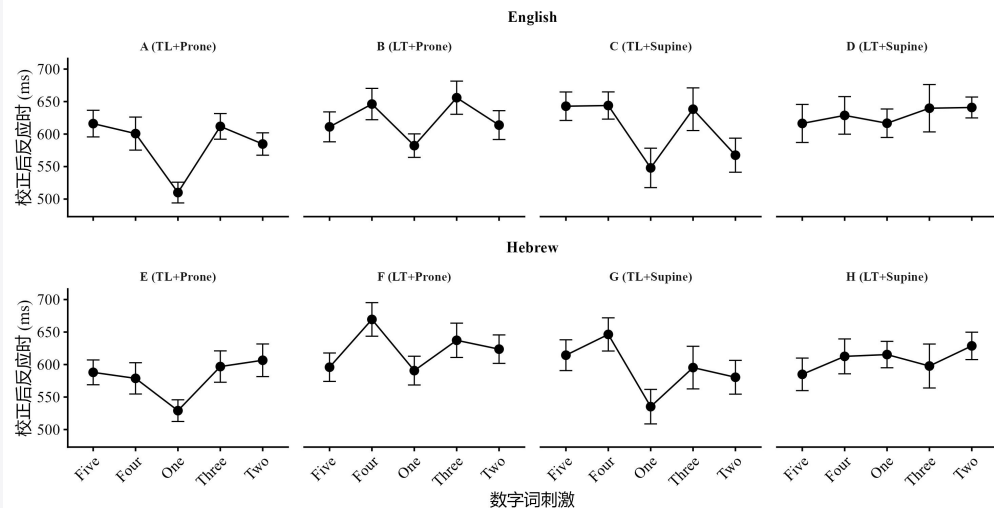
```
cat("\n--- Exp2: Mapping 柱状图 (Language 分面板) ---\n")
exp2_plot_dat <- exp2 %>%
  group_by(Mapping, Posture, Direction, Language, Subject) %>%
  summarise(RT = mean(RT), .groups = "drop") %>%
  group_by(Mapping, Posture, Direction, Language) %>%
  summarise(M = mean(RT), SE = sd(RT) / sqrt(n()), .groups = "drop") %>%
  mutate(Language = factor(Language, c("English", "Hebrew")),
         Condition = factor(paste0(Direction, "+", Posture),
                           c("TL+Prone", "LT+Prone", "TL+Supine", "LT+Supine")))

p2_main <- ggplot(exp2_plot_dat,
  aes(x = Mapping, y = M,
      fill = Condition)) +
  geom_col(position = position_dodge(0.7), width = 0.55,
    color = "black", linewidth = 0.4) +
  geom_errorbar(aes(ymin = M - SE, ymax = M + SE),
    position = position_dodge(0.7), width = 0.15, linewidth = 0.4) +
  facet_wrap(~ Language, ncol = 2, scales = "free_x") +
  scale_fill_manual(values = c("TL+Prone" = "white",
    "LT+Prone" = "grey60",
    "TL+Supine" = "grey85",
    "LT+Supine" = "grey30")) +
  labs(x = "Mapping", y = "校正后反应时 (ms)",
    title = "实验2: 数字词",
    fill = "条件") +
  theme_apa() +
  coord_cartesian(ylim = c(500, 700))
```



02 实验2对比——描述统计

实验2: 数字词效应



```
cat("\n--- Exp2: 刺激效应折线图 (Language x Mapping 分面板) ---\n")
exp2_stim_plot <- exp2 %>%
  group_by(Stimulus, Language, Mapping, Subject) %>%
  summarise(RT = mean(RT), .groups = "drop") %>%
  group_by(Stimulus, Language, Mapping) %>%
  summarise(M = mean(RT), SE = sd(RT) / sqrt(n()), .groups = "drop") %>%
  mutate(Language = factor(Language, c("English", "Hebrew")),
         Panel = case_when(
           Mapping == "A" ~ "A (TL+Prone)", Mapping == "B" ~ "B (LT+Prone)",
           Mapping == "C" ~ "C (TL+Supine)", Mapping == "D" ~ "D (LT+Supine)",
           Mapping == "E" ~ "E (TL+Prone)", Mapping == "F" ~ "F (LT+Prone)",
           Mapping == "G" ~ "G (TL+Supine)", Mapping == "H" ~ "H (LT+Supine)"),
         c("A (TL+Prone)", "B (LT+Prone)", "C (TL+Supine)", "D (LT+Supine)",
           "E (TL+Prone)", "F (LT+Prone)", "G (TL+Supine)", "H (LT+Supine)"))))

ylim2 <- range(exp2_stim_plot$M - exp2_stim_plot$SE,
               exp2_stim_plot$M + exp2_stim_plot$SE, na.rm = TRUE) + c(-10, 10)
p2_en <- exp2_stim_plot %>%
  filter(Language == "English") %>%
  ggplot(aes(x = Stimulus, y = M, group = 1)) +
  geom_line(linewidth = 0.5, color = "black") +
  geom_point(size = 2.5, color = "black") +
  geom_errorbar(aes(ymin = M - SE, ymax = M + SE),
                width = 0.3, linewidth = 0.4, color = "black") +
  facet_wrap(~ Panel, ncol = 4) +
  coord_cartesian(ylim = ylim2) +
  labs(subtitle = "English", x = NULL, y = "校正后反应时 (ms)") +
  theme_apo() +
  theme(strip.text = element_text(size = 9),
        plot.subtitle = element_text(hjust = 0.5, face = "bold", size = 11),
        axis.text.x = element_blank())

p2_he <- exp2_stim_plot %>%
  filter(Language == "Hebrew") %>%
  ggplot(aes(x = Stimulus, y = M, group = 1)) +
  geom_line(linewidth = 0.5, color = "black") +
  geom_point(size = 2.5, color = "black") +
  geom_errorbar(aes(ymin = M - SE, ymax = M + SE),
                width = 0.3, linewidth = 0.4, color = "black") +
  facet_wrap(~ Panel, ncol = 4) +
  coord_cartesian(ylim = ylim2) +
  labs(subtitle = "Hebrew", x = "数字词刺激", y = "校正后反应时 (ms)") +
  theme_apo() +
  theme(strip.text = element_text(size = 9),
        plot.subtitle = element_text(hjust = 0.5, face = "bold", size = 11),
        axis.text.x = element_text(angle = 45, hjust = 1))
p2_stim <- (p2_en / p2_he) +
  plot_annotation(title = "实验2: 数字词效应") &
  theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 13))
```

02 实验2对比——推断统计

```
cat("\n--- 2x2x2 RM-ANOVA: Direction x Posture x Language ---\n")
a2 <- aov_ez("Subject", "RT",
             exp2 %>% group_by(Subject, Direction, Posture, Language) %>%
             summarise(RT = mean(RT), .groups = "drop"),
             within = c("Direction", "Posture", "Language"),
             anova_table = list(es = "pes"))
print(a2)
```

基线任务 (2×5 RM-ANOVA : Posture×Finger)						
	<i>F</i>	df	<i>p</i>	η²p	原文显著性	推论
Posture	1.60	1, 20	.220	.074	significant	不一致
Finger	12.33	3.30, 65.95	<.001***	.381	significant	一致
Posture×Finger	4.54	3.10, 61.99	.006**	.185	no significant	不一致

```
b12_fit <- aov_ez("Subject", "RT", b12_agg,
                 within = c("Posture", "Finger"),
                 anova_table = list(es = "pes"))
print(b12_fit)
```

- 映射任务：
- Direction主效应和Direction×Posture交互作用复现成功，效应显著。
- 差异：Language主效应复现结果不显著但接近边缘 ($p=0.084$ vs. 原文： $p<0.001$, $\eta^2p=0.83$)。

2×2×2 RM-ANOVA: Direction×Posture×Language						
		<i>F</i>	df	<i>p</i>	η²p	
Direction	原文献	39.84	1, 20	<.001	.67	
	复现结果	37.55	1, 20	<.001	.652	
	推论			一致		
Posture	原文献			no significant		
	复现结果	0.21	1, 20	.653	.010	
	推论			一致		
Language	原文献	95.28	1, 20	<.001	.83	
	复现结果	3.31	1, 20	.084	.142	
	推论			不一致		
Direction×Posture	原文献	9.12	1, 20	<.01	.31	
	复现结果	7.64	1, 20	.012	.276	
	推论			一致		
Direction×Language	原文献			no significant		
	复现结果	0.00	1, 20	.971	<.001	
	推论			一致		
Posture×Language	原文献			no significant		
	复现结果	2.86	1, 20	.106	.125	
	推论			一致		
Direction×Posture×Language	原文献			no significant		
	复现结果	0.94	1, 20	.345	.045	
	推论			一致		

02 实验2对比——推断统计

One-way-ANOVA: Stimulus					
		F	df	p	η²p
English	原文献	35.54	4, 80	<.001	.64
	本研究	7.60	2.28, 45.58	<.001	.275
	推论			一致	
Hebrew	原文献	32.07	4, 80	<.001	.62
	本研究	5.60	2.64, 52.72	.003	.219
	推论			一致	

- 单因素方差分析：
 - 各语言内数字词主效应均复现显著，方向一致
 - 统计量和效应量均远小于原文献结果
- 事后比较：
 - 显著性复现成功率约为60%

```
## ---- 数字词刺激 ANOVA (各语言) ----
for (lang in c("English", "Hebrew")) {
  cat(sprintf("\n--- %s: 数字词 One-way RM-ANOVA ---\n", lang))
  d <- exp2 %>%
    filter(Language == lang) %>%
    group_by(Subject, Stimulus) %>%
    summarise(RT = mean(RT), .groups = "drop") %>%
    mutate(Stimulus = factor(Stimulus))

  a2d <- aov_ez("Subject", "RT", d, within = "Stimulus",
               anova_table = list(es = "pes"))

  print(a2d)
  cat("Scheffe Post-hoc:\n")
  print(pairs(emmeans(a2d, ~ Stimulus), adjust = "scheffe"))
}
```

contrast		estimate	t	p	原文献p值	结论
English	One-Two	-37.52	-2.318	0.289	< .001	不一致
	One-Three	-72.24	-4.181	0.011	< .001	一致
	One-Four	65.69	4.327	0.008	< .001	一致
	One-Five	57.40	6.589	< .001	< .001	一致
	Two-Three	34.72	1.771	0.549	< .05	不一致
	Two-Four	28.16	4.501	0.006	< .01	一致
	Two-Five	19.88	1.230	0.821		一致
	Three-Four	-6.56	-0.373	0.998		一致
	Three-Five	-14.84	-1.121	0.865	< .001	不一致
	Four-Five	-8.29	-0.592	0.985	< .005	不一致
Hebrew	One-Two	-42.26	-3.097	0.084	< .001	一致
	One-Three	-39.23	-2.329	0.284	< .001	不一致
	One-Four	59.18	3.874	0.020	< .001	一致
	One-Five	28.21	3.285	0.060	> .05	一致
	Two-Three	-3.04	-0.190	0.999		一致
	Two-Four	16.92	2.015	0.423		一致
	Two-Five	-14.05	-1.344	0.770	< .001	不一致
	Three-Four	19.96	1.546	0.669		一致
	Three-Five	-11.01	-0.787	0.959	< .001	不一致
	Four-Five	-30.97	-2.627	0.184	< .001	不一致

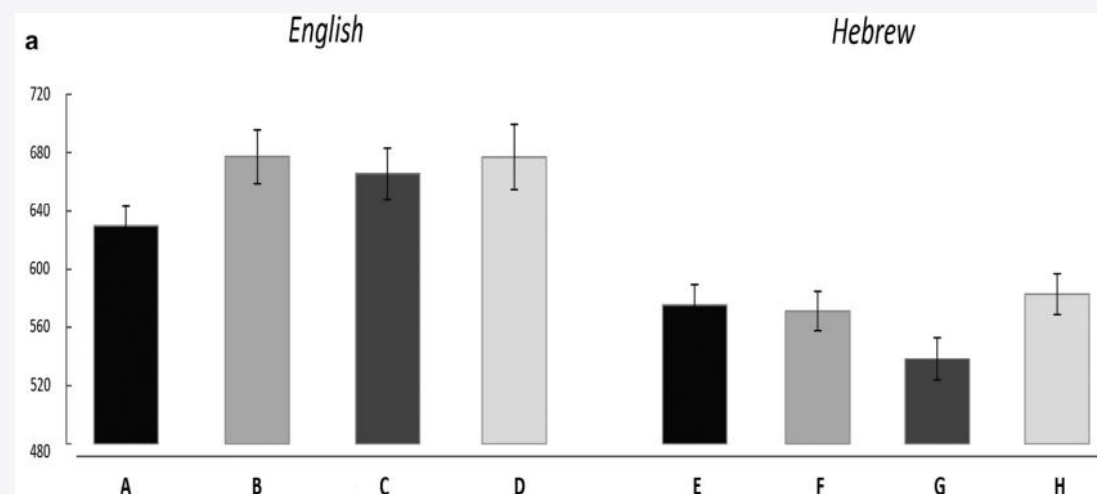
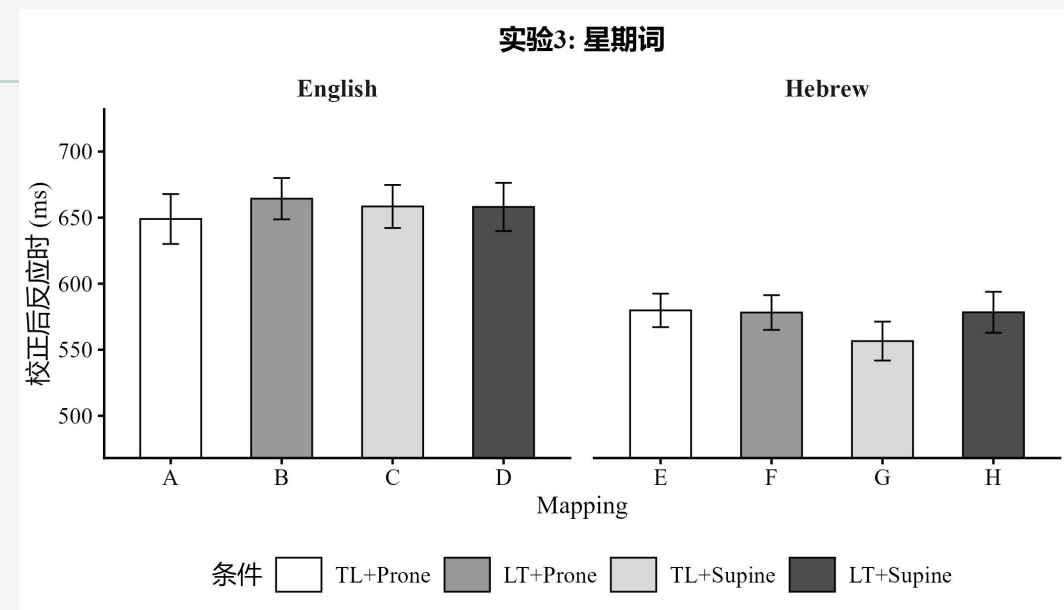
02 实验3对比——描述统计

		原文献结果		复现结果		M_δ	M_评级	SD_δ	SD_评级
		M	SD	M	SD				
Direction	TL	602.3	82.2	610.93	113.53	1.41%	次要偏差	27.60%	主要偏差
	LT	627	91.6	619.74	110.07	1.17%	次要偏差	16.78%	主要偏差
Language	English	662.4	80.7	657.46	116.22	0.75%	次要偏差	30.56%	主要偏差
	Hebrew	566.9	62.3	573.20	89.32	1.10%	次要偏差	30.25%	主要偏差
Mapping	A	629.6	61.2	648.95	121.35	2.98%	次要偏差	49.57%	主要偏差
	B	677.3	82.3	664.35	101.67	1.95%	次要偏差	19.05%	主要偏差
	C	665.6	79	658.44	117.84	1.09%	次要偏差	32.96%	主要偏差
	D	676.9	100.2	658.10	123.97	2.86%	次要偏差	19.17%	主要偏差
	E	575.6	61.7	579.78	78.69	0.72%	次要偏差	21.59%	主要偏差
	F	571.1	60	578.16	79.49	1.22%	次要偏差	24.52%	主要偏差
	G	538.5	65.3	556.54	97.11	3.24%	次要偏差	32.76%	主要偏差
	H	582.8	62.2	578.34	99.16	0.77%	次要偏差	37.27%	主要偏差
Days - English	Sun	584.17	56.89	585.92	54.95	0.30%	次要偏差	3.53%	次要偏差
	Mon	608.41	58.02	610.09	57.80	0.28%	次要偏差	0.38%	次要偏差
	Tues	723.57	101.57	716.23	112.83	1.02%	次要偏差	9.98%	次要偏差
	Wed	661.83	76.61	648.74	80.38	2.02%	次要偏差	4.69%	次要偏差
Days - Hebrew	Thurs	733.79	110.2	726.31	100.70	1.03%	次要偏差	9.43%	次要偏差
	Sun	528.61	57.24	530.25	51.73	0.31%	次要偏差	10.65%	主要偏差
	Mon	555.99	57.92	562.41	58.16	1.14%	次要偏差	0.41%	次要偏差
	Tues	583.93	65.84	590.20	75.80	1.06%	次要偏差	13.14%	主要偏差
	Wed	590.99	64.98	600.57	68.95	1.60%	次要偏差	5.76%	次要偏差
	Thurs	575.41	56.64	582.60	58.54	1.23%	次要偏差	3.25%	次要偏差

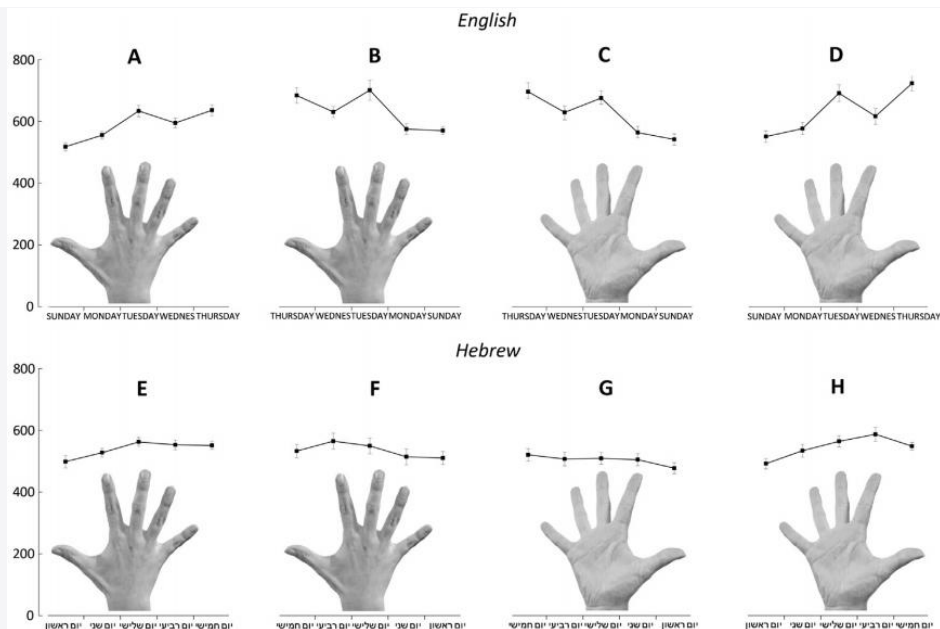
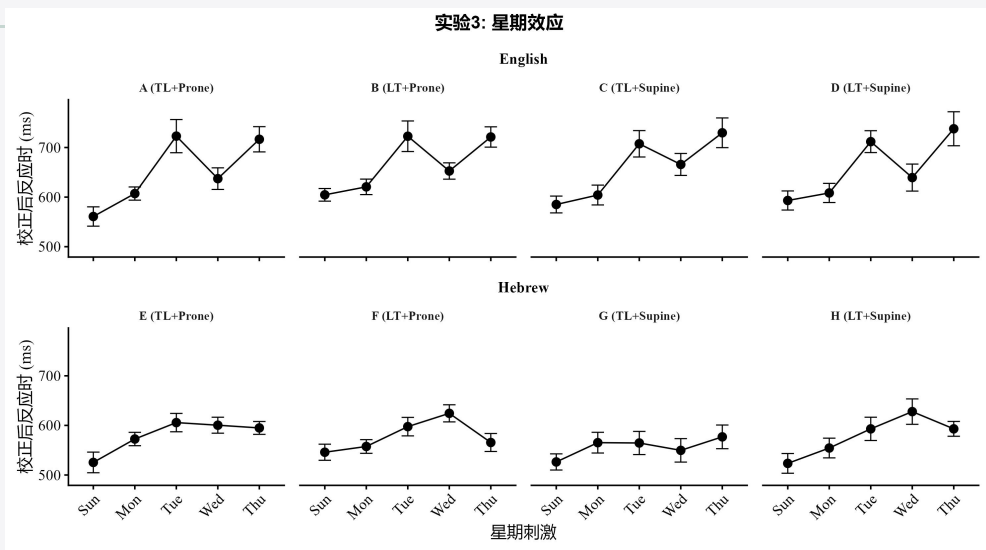
02 实验3对比——描述统计

```
cat("\n--- Exp3: Mapping 柱状图 (Language 分面板) ---\n")
exp3_plot_dat <- exp3 %>%
  group_by(Mapping, Posture, Direction, Language, Subject) %>%
  summarise(RT = mean(RT), .groups = "drop") %>%
  group_by(Mapping, Posture, Direction, Language) %>%
  summarise(M = mean(RT), SE = sd(RT) / sqrt(n()), .groups = "drop") %>%
  mutate(Language = factor(Language, c("English", "Hebrew")),
         Condition = factor(paste0(Direction, "+", Posture),
                           c("TL+Prone", "LT+Prone", "TL+Supine", "LT+Supine")))

p3_main <- ggplot(exp3_plot_dat,
  aes(x = Mapping, y = M,
      fill = Condition)) +
  geom_col(position = position_dodge(0.7), width = 0.55,
    color = "black", linewidth = 0.4) +
  geom_errorbar(aes(ymin = M - SE, ymax = M + SE),
    position = position_dodge(0.7), width = 0.15, linewidth = 0.4) +
  facet_wrap(~ Language, ncol = 2, scales = "free_x") +
  scale_fill_manual(values = c("TL+Prone" = "white",
    "LT+Prone" = "grey60",
    "TL+Supine" = "grey85",
    "LT+Supine" = "grey30")) +
  labs(x = "Mapping", y = "校正后反应时 (ms)",
    title = "实验3: 星期词",
    fill = "条件") +
  theme_apo() +
  coord_cartesian(ylim = c(480, 720))
```



02 实验3对比——描述统计



```
cat("\n--- Exp3: 刺激效应折线图 (Language x Mapping 分面板) ---\n")
exp3_stim_plot <- exp3 %>%
  mutate(Language = factor(Language, c("English", "Hebrew")),
         Stimulus = factor(Stimulus,
                           c("Sunday", "Monday", "Tuesday", "Wednesday", "Thursday"))) %>%
  group_by(Stimulus, Language, Mapping, Subject) %>%
  summarise(RT = mean(RT), .groups = "drop") %>%
  group_by(Stimulus, Language, Mapping) %>%
  summarise(M = mean(RT), SE = sd(RT) / sqrt(n()), .groups = "drop") %>%
  mutate(Panel = case_when(
    Mapping == "A" ~ "A (TL+Prone)", Mapping == "B" ~ "B (LT+Prone)",
    Mapping == "C" ~ "C (TL+Supine)", Mapping == "D" ~ "D (LT+Supine)",
    Mapping == "E" ~ "E (TL+Prone)", Mapping == "F" ~ "F (LT+Prone)",
    Mapping == "G" ~ "G (TL+Supine)", Mapping == "H" ~ "H (LT+Supine)",
    Panel = factor(Panel,
                  c("A (TL+Prone)", "B (LT+Prone)", "C (TL+Supine)", "D (LT+Supine)",
                    "E (TL+Prone)", "F (LT+Prone)", "G (TL+Supine)", "H (LT+Supine)"))))

day_labels <- c("Sun", "Mon", "Tue", "Wed", "Thu")
ylim3 <- range(exp3_stim_plot$M - exp3_stim_plot$SE,
               exp3_stim_plot$M + exp3_stim_plot$SE, na.rm = TRUE) + c(-10, 10)
p3_en <- exp3_stim_plot %>%
  filter(Language == "English") %>%
  ggplot(aes(x = Stimulus, y = M, group = 1)) +
  geom_line(linewidth = 0.5, color = "black") +
  geom_point(size = 2.5, color = "black") +
  geom_errorbar(aes(ymin = M - SE, ymax = M + SE),
                width = 0.3, linewidth = 0.4, color = "black") +
  scale_x_discrete(labels = day_labels) +
  facet_wrap(~ Panel, ncol = 4) +
  coord_cartesian(ylim = ylim3) +
  labs(subtitle = "English", x = NULL, y = "校正后反应时 (ms)") +
  theme_apa() +
  theme(strip.text = element_text(size = 9),
        plot.subtitle = element_text(hjust = 0.5, face = "bold", size = 11),
        axis.text.x = element_blank())
p3_he <- exp3_stim_plot %>%
  filter(Language == "Hebrew") %>%
  ggplot(aes(x = Stimulus, y = M, group = 1)) +
  geom_line(linewidth = 0.5, color = "black") +
  geom_point(size = 2.5, color = "black") +
  geom_errorbar(aes(ymin = M - SE, ymax = M + SE),
                width = 0.3, linewidth = 0.4, color = "black") +
  scale_x_discrete(labels = day_labels) +
  facet_wrap(~ Panel, ncol = 4) +
  coord_cartesian(ylim = ylim3) +
  labs(subtitle = "Hebrew", x = "星期刺激", y = "校正后反应时 (ms)") +
  theme_apa() +
  theme(strip.text = element_text(size = 9),
        plot.subtitle = element_text(hjust = 0.5, face = "bold", size = 11),
        axis.text.x = element_text(angle = 45, hjust = 1))
p3_stim <- (p3_en / p3_he) +
  plot_annotation(title = "实验3: 星期效应") &
  theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 13))
```

02 实验3对比——推断统计

```
cat("\n--- 2x2x2 RM-ANOVA: Direction x Posture x Language ---\n")
a3 <- aov_ez("Subject", "RT",
             exp3 %>% group_by(Subject, Direction, Posture, Language) %>%
             summarise(RT = mean(RT), .groups = "drop"),
             within = c("Direction", "Posture", "Language"),
             anova_table = list(es = "pes"))
print(a3)
```

基线任务 (2×5 RM-ANOVA: Posture×Finger)						
	<i>F</i>	df	<i>p</i>	η² _p	原文显著性	推论
Posture	27.55	1, 19	<.001	.592	significant	一致
Finger	7.68	3.12, 59.25	<.001	.288	significant	一致
Posture×Finger	1.50	3.16, 60.00	.222	.073	no significant	一致

```
b13_fit <- aov_ez("Subject", "RT", b13_agg,
                 within = c("Posture", "Finger"),
                 anova_table = list(es = "pes"))
print(b13_fit)
```

- 映射任务：
 - 复现一致：Language主效应显著；三重交互效应显著，效应量偏差较小，印证原文核心发现。
 - 差异：原文Direction主效应显著，复现结果不显著但接近边缘 ($p=0.084$)。

手指映射任务 (2×2×2 RM-ANOVA: Direction×Posture×Language)					
		<i>F</i>	df	<i>p</i>	η² _p
Direction	原文献	34.94	1, 19	< .001	.65
	复现结果	3.32	1, 19	.084	.149
	推论			不一致	
Posture	原文献			no significant	
	复现结果	0.34	1, 19	.566	.018
	推论			一致	
Language	原文献	102.19	1, 19	< .001	.84
	复现结果	112.95	1, 19	< .001	.856
	推论			一致	
Direction×Posture	原文献			no significant	
	复现结果	0.18	1, 19	.672	.010
	推论			一致	
Direction×Language	原文献			no significant	
	复现结果	0.10	1, 19	.751	.005
	推论			一致	
Posture×Language	原文献			no significant	
	复现结果	0.84	1, 19	.370	.042
	推论			一致	
Direction×Posture×Language	原文献	16.69	1, 19	< .002	.47
	复现结果	8.54	1, 19	.009	.310
	推论			一致	

02 实验3对比——推断统计

```
## ---- 星期刺激 ANOVA (各语言) ----
for (lang in c("English", "Hebrew")) {
  cat(sprintf("\n--- %s: 星期 One-way RM-ANOVA ---\n", lang))
  d <- exp3 %>%
    filter(Language == lang) %>%
    group_by(Subject, Stimulus) %>%
    summarise(RT = mean(RT), .groups = "drop") %>%
    mutate(Stimulus = factor(Stimulus, c("Sunday", "Monday", "Tuesday", "Wednesday", "Thursday")))
  a3d <- aov_ez("Subject", "RT", d, within = "Stimulus",
    anova_table = list(es = "pes"))
  print(a3d)
  cat("Scheffe Post-hoc:\n")
  print(pairs(emmeans(a3d, ~ Stimulus), adjust = "scheffe"))
}
```

One-way-ANOVA: Stimulus					
		<i>F</i>	df	<i>p</i>	η^2p
English	原文献	33.97	4, 76	< .001	.64
	复现结果	27.57	2.14, 40.57	<.001	.592
	推论			一致	
Hebrew	原文献	14.62	4, 76	< .001	.44
	复现结果	14.57	2.59, 49.28	< .001	.434
	推论			一致	

- 单因素方差分析：
 - 各语言内星期词主效应均成功复现
 - 方向与原文献一致，效应量偏差较小

contrast		estimate	<i>t</i>	<i>p</i>	原文献 <i>p</i> 值	结论
English	Sun - Mon	-24.2	-2.434	0.247		一致
	Sun - Tues	-130.3	-5.518	0.001	< .001	一致
	Sun - Wed	-62.8	-4.189	0.011	< .001	一致
	Sun - Thurs	-140.4	-7.604	< .001	< .001	一致
	Mon - Tues	-106.1	-5.322	0.001	< .001	一致
	Mon - Wed	-38.7	-4.484	0.006	< .001	一致
	Mon - Thurs	-116.2	-6.151	< .001	< .001	一致
	Tues - Wed	67.5	4.235	0.010	< .005	一致
	Tues -Thurs	-10.1	-0.682	0.975		一致
	Wed - Thurs	-77.6	-4.397	0.007	< .05	一致
Hebrew	Sun - Mon	-32.17	-3.861	0.021	< .05	一致
	Sun - Tues	-59.95	-5.152	0.002	< .001	一致
	Sun - Wed	-70.32	-7.381	< .001	< .001	一致
	Sun - Thurs	-52.36	-7.525	< .001	< .001	一致
	Mon - Tues	-27.79	-2.006	0.429		一致
	Mon - Wed	-38.15	-6.290	< .001	< .005	一致
	Mon - Thurs	-20.19	-2.039	0.413		一致
	Tues - Wed	-10.37	-0.944	0.922		一致
	Tues -Thurs	7.59	0.582	0.986		一致
	Wed - Thurs	17.96	1.813	0.527		一致

- 事后比较：显著性复现成功率约为100%

02 线性混合模型(LMM)复现

	固定效应	<i>F</i>	df	<i>p</i>	原文献报告	RM-ANOVA复现
实验1: corrected_RT ~	Direction	42.31	3347.1	< .001 ***	F(1,17) = 32.65, <i>p</i> < .001	F(1,17) = 12.25, <i>p</i> = .003
Direction * Posture + (1 Subject)	Posture	20.66	3347.0	< .001 ***	no significant	F(1,17) = 5.91, <i>p</i> = .026
	Direction × Posture	0.80	3347.0	.370	no significant	F(1,17) = 0.21, <i>p</i> = .651
实验2:	Direction	68.76	7726.0	< .001 ***	F(1,20) = 39.84, <i>p</i> < .001	F(1,20) = 37.55, <i>p</i> < .001
corrected_RT ~ Direction *	Posture	6.60	7726.1	.010*	no significant	F(1,17) = 0.21, <i>p</i> = .653
Posture * Language + (1 Subject)	Language	10.88	7726.0	< .001 ***	F(1,20) = 95.28, <i>p</i> < .001	F(1,20) = 3.31, <i>p</i> = .084
	Direction × Posture	10.93	7726.0	< .001 ***	F(1,20) = 9.12, <i>p</i> < .01	F(1,20) = 7.64, <i>p</i> = .012
	Direction × Language	0.03	7726.0	.861	no significant	F(1,20) < 0.01, <i>p</i> = .971
	Posture × Language	6.04	7726.0	.014*	no significant	F(1,20) = 2.86, <i>p</i> = .106
	Direction × Posture × Language	0.32	7726.0	.570	no significant	F(1,17) = 0.94, <i>p</i> = .345
实验3:	Direction	3.87	7379.1	.049*	F(1,19) = 34.94, <i>p</i> < .001	F(1,19) = 3.32, <i>p</i> = .084
corrected_RT ~ Direction *	Posture	4.71	7379.1	.030*	no significant	F(1,19) = 0.34, <i>p</i> = .566
Posture * Language + (1 Subject)	Language	621.73	7379.1	< .001***	F(1,19) = 102.19, <i>p</i> < .001	F(1,19) = 112.95, <i>p</i> < .001
	Direction × Posture	0.05	7379.1	.818	no significant	F(1,19) = 0.18, <i>p</i> = .672
	Direction × Language	1.11	7379.1	.292	no significant	F(1,19) = 0.10, <i>p</i> = .751
	Posture × Language	3.47	7379.1	.063	no significant	F(1,19) = 0.84, <i>p</i> = .370
	Direction × Posture × Language	10.91	7379.0	.001**	F(1,19) = 16.69, <i>p</i> < .002	F(1,19) = 8.54, <i>p</i> = .009

02 整体可复现性

核心理论结论：方向性成立，基本可复现

- 数字序列加工由 FC 机制主导 (实验1、2)
- 非数字序列则由 FC 与 RD 机制协同作用 (实验3)

精细统计指标：可靠性存疑，需审慎解读

- 复现的效应量 (F 值、 η^2p) 普遍小于原报告
- 实验2的Language主效应对统计方法敏感
- 实验1、2中部分事后比较的显著性无法稳定复现

结果可复现性的评估表

可复现性情况	数量及占比	
	<i>N</i>	%
完全一致($\delta = 0\%$)	20	23.26%
偏差较小($0\% < \delta < 10\%$)	51	59.30%
偏差较大($\delta > 10\%$)	15	17.44%
因舍入导致的偏差	0	0%

推论一致性的评估表 (RM-ANOVA)

推论一致性	数量及占比	
	<i>N</i>	%
一致	75	76.53%
不一致	23	23.47%

02 偏差来源分析

校正步骤不透明

- 手指校正和姿势校：具体校正公式的细节在文中未完全公开（如扣减顺序）
- 两步RT校正会叠加舍入误差，放大偏差，影响结果精度

异常值处理方法差异

- 文献表述不清：overall or individual；基线与映射任务均剔除异常值？
- 异常值处理的顺序未知：校正前 or 校正后

软件与统计方法差异

- SPSS vs. R：Greenhouse-Geisser校正的 ϵ 系数计算、自由度舍入方式等方面可能存在细微差异
- 传统RM-ANOVA vs. LMM：实验2的Language主效应（RM-ANOVA: 不显著 vs. LMM: 显著）

THANK YOU!

